Maths, one drawing at a time

Maths Board: Syllabus

From the core pictures to CSS General Ability reasoning, in 9 sections

9	50	309	163
sections	lessons	board steps	questions

How every lesson works

1. A picture of the idea, drawn step by step on the board.
2. Worked examples, one line at a time, each line explained.
3. A "why not the other way?" board: the common mistake in red, the right way in green.
4. A short "Remember" summary.
5. Check questions inside the lesson and practice questions at the end, with a reason for every answer.

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Contents

Start here: the core pictures

1 Numbers live on a line

2 The four operations as pictures

Number

3 Place value: what each digit is worth

4 Adding and taking away in columns

5 Multiplying bigger numbers

6 Dividing and remainders

7 BODMAS: which part first?

8 Minus numbers: adding, taking away and times

9 Factors, multiples and primes

10 HCF and LCM

11 Even, odd, consecutive numbers and remainders

Fractions, decimals, percentages & money

12 What a fraction is, and equal fractions

13 Adding fractions with different denominators

14 Multiplying and dividing fractions

15 Decimals and rounding

16 Percentages: what they mean

17 Percentage increase, decrease and reverse

18 Ratio and sharing

19 Direct and inverse proportion

20 Profit, loss and discount

21 Simple and compound interest

Algebra

22 Letters that stand for numbers

23 Solving equations: keep it balanced

24 Turning stories into equations

25 Two unknowns at once

Geometry

26 Angles: turns, lines and points

27 Parallel lines and the angles they make

28 Triangles

29	Polygons and their angles
30	Perimeter and area
31	Circles
32	Volume of cubes and cuboids
33	Pythagoras

Graphs

34	Coordinates: finding points on a grid
35	Reading bar charts and tables

Statistics & probability

36	The mean (average)
37	Median and mode
38	Sets and Venn diagrams
39	Population, sample and fair sampling
40	Probability: how likely?

Advanced: speed, work & rates

41	Speed, distance and time
42	Average speed and relative speed
43	Working together
44	Pipes and tanks

Reasoning: patterns, logic & puzzles

45	Number and letter series
46	Coding and decoding
47	All, some and none: syllogisms
48	If-then, necessary and sufficient
49	Arrangement puzzles
50	Directions, clocks and gears

Start here: the core pictures

1. Numbers live on a line

Every number has a place on a line. Adding moves right, taking away moves left.

YOU WILL LEARN TO

- Put numbers in order on a number line.
- Add and take away by jumping along the line.
- See minus numbers to the left of zero.

BOARD STEPS

1. A number line
2. Before zero: minus numbers
3. Worked example 1
4. Worked example 2
5. Worked example 3
6. Why not the other way?
7. Remember

7 steps · 1 check question · 3 practice questions

2. The four operations as pictures

Plus joins, minus takes away, times makes equal groups, divide shares fairly.

YOU WILL LEARN TO

- See + as joining and – as taking away.
- See × as equal groups.
- See ÷ as fair sharing, and know which sums can be swapped.

BOARD STEPS

1. Add = join, take away = remove
2. Times = equal groups, divide = fair sharing
3. Worked example 1
4. Worked example 2
5. Worked example 3
6. Why not the other way?
7. Remember

7 steps · 1 check question · 3 practice questions

Number

3. Place value: what each digit is worth

A digit is worth more the further left it sits: each place is 10 times the one on its right.

YOU WILL LEARN TO

- Read the value of each digit.
- Write numbers in expanded form.
- Compare and round whole numbers.

BOARD STEPS

1. Each place is worth 10 times more
2. Worked example 1
3. Worked example 2
4. Worked example 3
5. Why not the other way?
6. Remember

6 steps · 1 check question · 3 practice questions

4. Adding and taking away in columns

Line up the places, start from the ones, carry when a column reaches 10, borrow when the top is too small.

YOU WILL LEARN TO

- Add big numbers in columns, carrying when needed.
- Take away in columns, borrowing when needed.
- Check an answer by doing the opposite sum.

BOARD STEPS

1. Adding with carrying: 487 + 256
2. Taking away with borrowing: 503 – 178
3. Worked example 1
4. Worked example 2
5. Worked example 3
6. Why not the other way?
7. Remember

7 steps · 0 check questions · 3 practice questions

5. Multiplying bigger numbers

Split a number into tens and ones, multiply each part, then add the parts.

YOU WILL LEARN TO

- See multiplication as a rectangle of equal rows.
- Multiply 2-digit numbers by splitting them.
- Use multiplying in word problems.

BOARD STEPS

1. 23 × 14 as a rectangle
2. Worked example 1
3. Worked example 2
4. Worked example 3
5. Why not the other way?
6. Remember

6 steps · 0 check questions · 3 practice questions

6. Dividing and remainders

Divide by sharing into equal groups. What is left over is the remainder, and it is always smaller than the divisor.

YOU WILL LEARN TO

- Divide by finding the biggest multiple that fits.
- Find and check the remainder.
- Decide what to do with a remainder in a word problem.

BOARD STEPS

1. 14 sweets shared by 3 children
2. Worked example 1
3. Worked example 2
4. Worked example 3
5. Why not the other way?
6. Remember

6 steps · 0 check questions · 3 practice questions

7. BODMAS: which part first?

Brackets, Orders, Divide and Multiply, Add and Subtract: always in this order.

YOU WILL LEARN TO

- Know the order of operations.
- Work out × and ÷ (or + and –) from left to right.
- Avoid the most common BODMAS traps.

BOARD STEPS

1. The order, from first to last
2. Worked example 1
3. Worked example 2
4. Worked example 3
5. Why not the other way?
6. Remember

6 steps · 0 check questions · 3 practice questions

8. Minus numbers: adding, taking away and times

Use the number line for + and –, and the sign rule for × and ÷.

YOU WILL LEARN TO

- Add and take away with minus numbers.
- Understand why taking away a minus adds.
- Multiply and divide with signs.

BOARD STEPS

1. Moving on the number line
2. Signs when multiplying
3. Worked example 1
4. Worked example 2
5. Worked example 3
6. Why not the other way?
7. Remember

7 steps · 1 check question · 3 practice questions

9. Factors, multiples and primes

Factors go into a number exactly. Multiples are in its times table. Primes have exactly two factors.

YOU WILL LEARN TO

- Find all the factors of a number.
- Tell factors from multiples.
- Spot prime numbers and use divisibility tests.

BOARD STEPS

1. 12 counters in rectangles
2. Prime numbers
3. Worked example 1
4. Worked example 2
5. Worked example 3
6. Why not the other way?
7. Remember

7 steps · 1 check question · 3 practice questions

10. HCF and LCM

HCF is the biggest number that divides both. LCM is the smallest number both divide into.

YOU WILL LEARN TO

- Find the HCF and LCM by listing.
- Know which one a word problem needs.
- Check with HCF × LCM = the product.

BOARD STEPS

1. Shared factors and shared multiples
2. Worked example 1
3. Worked example 2
4. Worked example 3
5. Why not the other way?
6. Remember

6 steps · 0 check questions · 3 practice questions

11. Even, odd, consecutive numbers and remainders

Even numbers make pairs, odd numbers have one left over. Three numbers in a row add to three times the middle one.

YOU WILL LEARN TO

- Use even and odd rules.
- Solve consecutive-number problems.
- Find remainders without big division.

BOARD STEPS

1. Even and odd as pairs
2. Worked example 1
3. Worked example 2
4. Worked example 3
5. Why not the other way?
6. Remember

6 steps · 0 check questions · 3 practice questions

Fractions, decimals, percentages & money

12. What a fraction is, and equal fractions

A fraction is equal parts of one whole. Multiplying or dividing the top and bottom by the same number keeps its value.

YOU WILL LEARN TO

- Read the top and bottom of a fraction.
- Make equivalent fractions and simplify.
- Compare fractions and write mixed numbers.

BOARD STEPS

1. One half, three ways
2. Worked example 1
3. Worked example 2
4. Worked example 3
5. Why not the other way?
6. Remember

6 steps · 1 check question · 3 practice questions

13. Adding fractions with different denominators

How to add fractions like 1/2 + 1/3, where the pieces are different sizes.

YOU WILL LEARN TO

- Say what the top and bottom numbers of a fraction mean.
- Explain why pieces of different sizes cannot be counted together.
- Cut both fractions into same-size pieces (a common denominator).
- Add the tops, keep the bottom, and check the answer makes sense.

BOARD STEPS

1. A fraction is a whole cut into equal parts
2. The question, and why it is tricky
3. Cut both into same-size pieces
4. Now the pieces match: count them
5. The same method on three examples
6. Why not just add the bottoms?
7. Which answers fit?
8. Remember the method

8 steps · 5 check questions · 3 practice questions

14. Multiplying and dividing fractions

Multiply tops and bottoms. To divide, flip the second fraction and multiply.

YOU WILL LEARN TO

- Multiply fractions and see why it works.
- Understand "of" as multiply.
- Divide by a fraction using keep, change, flip.

BOARD STEPS

1. Half of a third
2. Worked example 1
3. Worked example 2
4. Worked example 3
5. Why not the other way?
6. Remember

6 steps · 0 check questions · 3 practice questions

15. Decimals and rounding

Digits after the point are tenths, hundredths and thousandths. Line up points to add; count decimal places to multiply.

YOU WILL LEARN TO

- Read decimal place values.
- Add, take away and multiply decimals.
- Change fractions to decimals and round numbers.

BOARD STEPS

1. Tenths and hundredths
2. Worked example 1
3. Worked example 2
4. Worked example 3
5. Why not the other way?
6. Remember

6 steps · 0 check questions · 3 practice questions

16. Percentages: what they mean

Percent means out of 100. The amount you start with is always 100%.

YOU WILL LEARN TO

- Change between percentages, fractions and decimals.
- Find a percentage of an amount.
- Write one number as a percentage of another.

BOARD STEPS

1. 25 out of 100
2. Worked example 1
3. Worked example 2
4. Worked example 3
5. Why not the other way?
6. Remember

6 steps · 0 check questions · 3 practice questions

17. Percentage increase, decrease and reverse

Change ÷ the OLD amount × 100. For reverse questions, the new amount is (100 ± p)% of the old one.

YOU WILL LEARN TO

- Find a percentage increase or decrease.
- Find the original amount (reverse percentage).
- Handle two changes in a row.

BOARD STEPS

1. An increase of 20%
2. Worked example 1
3. Worked example 2
4. Worked example 3
5. Why not the other way?
6. Remember

6 steps · 0 check questions · 3 practice questions

18. Ratio and sharing

A ratio counts parts. Add the parts, find one part, then multiply.

YOU WILL LEARN TO

- Read and simplify ratios.
- Share an amount in a ratio.
- Solve the lottery-sharing and joined-ratio questions.

BOARD STEPS

1. Sharing Rs 1,500 in the ratio 2 : 3
2. Worked example 1
3. Worked example 2
4. Worked example 3
5. Why not the other way?
6. Remember

6 steps · 0 check questions · 3 practice questions

19. Direct and inverse proportion

Direct: both go up together, so find the value of one. Inverse: one goes up as the other goes down, so the product stays the same.

YOU WILL LEARN TO

- Use the unitary method.
- Spot direct and inverse proportion.
- Solve worker-days problems.

BOARD STEPS

1. Two kinds of proportion
2. Worked example 1
3. Worked example 2
4. Worked example 3
5. Why not the other way?
6. Remember

6 steps · 0 check questions · 3 practice questions

20. Profit, loss and discount

Profit and loss percentages are on the cost price. Discount is on the marked price.

YOU WILL LEARN TO

- Find profit or loss and its percentage.
- Work out a discount and the price paid.
- Find the cost price from the selling price.

BOARD STEPS

1. From cost to selling price
2. Worked example 1
3. Worked example 2
4. Worked example 3
5. Why not the other way?
6. Remember

6 steps · 0 check questions · 3 practice questions

21. Simple and compound interest

Simple interest is the same every year. Compound interest is paid on the interest too, so it grows faster.

YOU WILL LEARN TO

- Use I = P × R × T ÷ 100.
- Work out compound interest year by year.
- Compare simple and compound interest.

BOARD STEPS

1. Rs 10,000 at 10% for 2 years
2. Worked example 1
3. Worked example 2
4. Worked example 3
5. Why not the other way?
6. Remember

6 steps · 0 check questions · 3 practice questions

Algebra

22. Letters that stand for numbers

A letter is a box that hides a number. Put numbers in, and only add terms that are alike.

YOU WILL LEARN TO

- Read expressions like 3x + 5.
- Substitute numbers for letters.
- Collect like terms.

BOARD STEPS

1. A letter is a box that hides a number
2. Worked example 1
3. Worked example 2
4. Worked example 3
5. Why not the other way?
6. Remember

6 steps · 0 check questions · 3 practice questions

23. Solving equations: keep it balanced

An equation is a balance. Do the same thing to both sides until the letter is alone.

YOU WILL LEARN TO

- See an equation as a balance.
- Solve one-step and two-step equations.
- Solve equations with brackets and letters on both sides.

BOARD STEPS

1. An equation is a balance
2. Worked example 1
3. Worked example 2
4. Worked example 3
5. Why not the other way?
6. Remember

6 steps · 0 check questions · 3 practice questions

24. Turning stories into equations

Name the unknown, translate each sentence, solve, then check in the story.

YOU WILL LEARN TO

- Choose a letter for the unknown.
- Translate sentences into an equation.
- Solve number and age problems.

BOARD STEPS

1. From words to an equation
2. Worked example 1
3. Worked example 2
4. Worked example 3
5. Why not the other way?
6. Remember

6 steps · 0 check questions · 3 practice questions

25. Two unknowns at once

With two unknowns you need two facts. Add or subtract the equations so one letter disappears.

YOU WILL LEARN TO

- Write two equations from a story.
- Remove one letter by adding or subtracting.
- Find both unknowns and check.

BOARD STEPS

1. Make one letter disappear
2. Worked example 1
3. Worked example 2
4. Worked example 3
5. Why not the other way?
6. Remember

6 steps · 0 check questions · 3 practice questions

Geometry

26. Angles: turns, lines and points

Angles on a straight line add to 180°, around a point to 360°, and vertically opposite angles are equal.

YOU WILL LEARN TO

- Name acute, right, obtuse, straight and reflex angles.
- Use straight-line and around-a-point facts.
- Use complementary and supplementary angles.

BOARD STEPS

1. Kinds of angle
2. Angle facts
3. Worked example 1
4. Worked example 2
5. Worked example 3
6. Why not the other way?
7. Remember

7 steps · 1 check question · 3 practice questions

27. Parallel lines and the angles they make

When a line crosses two parallel lines: corresponding angles are equal, alternate angles are equal, co-interior angles add to 180°.

YOU WILL LEARN TO

- Recognise parallel lines and a transversal.
- Find corresponding and alternate angles.
- Use co-interior angles.

BOARD STEPS

1. F, Z and C shapes
2. Worked example 1
3. Worked example 2
4. Worked example 3
5. Why not the other way?
6. Remember

6 steps · 0 check questions · 3 practice questions

28. Triangles

The angles of every triangle add up to 180°. Isosceles triangles have two equal sides and two equal angles.

YOU WILL LEARN TO

- Use the angle sum of a triangle.
- Know equilateral, isosceles and scalene triangles.
- Use the exterior angle fact.

BOARD STEPS

1. Why a triangle's angles make 180°
2. Three kinds of triangle
3. Worked example 1
4. Worked example 2
5. Worked example 3
6. Why not the other way?
7. Remember

7 steps · 1 check question · 3 practice questions

29. Polygons and their angles

An n-sided polygon splits into n – 2 triangles, so its angles add to (n – 2) × 180°. Exterior angles always add to 360°.

YOU WILL LEARN TO

- Find the interior angle sum of any polygon.
- Find each angle of a regular polygon.
- Use exterior angles to find the number of sides.

BOARD STEPS

1. Split into triangles
2. Worked example 1
3. Worked example 2
4. Worked example 3
5. Why not the other way?
6. Remember

6 steps · 0 check questions · 3 practice questions

30. Perimeter and area

Perimeter is the distance around. Area is how many unit squares cover the inside.

YOU WILL LEARN TO

- Find perimeters of shapes.
- Find areas of rectangles, triangles and parallelograms.
- Choose the right one in word problems.

BOARD STEPS

1. Around the edge, or the inside?
2. Worked example 1
3. Worked example 2
4. Worked example 3
5. Why not the other way?
6. Remember

6 steps · 0 check questions · 3 practice questions

31. Circles

Circumference = $2\pi r$, area = πr^2. The diameter is twice the radius.

YOU WILL LEARN TO

- Know radius, diameter and circumference.
- Find the circumference of a circle.
- Find the area of a circle.

BOARD STEPS

1. Parts of a circle
2. Worked example 1
3. Worked example 2
4. Worked example 3
5. Why not the other way?
6. Remember

6 steps · 0 check questions · 3 practice questions

32. Volume of cubes and cuboids

Volume = length × width × height, measured in cubic units. 1 litre = 1000 cm^3.

YOU WILL LEARN TO

- Find the volume of a cuboid and a cube.
- Use cubic units and litres.
- Solve packing and tank problems.

BOARD STEPS

1. Counting cubes
2. Worked example 1
3. Worked example 2
4. Worked example 3
5. Why not the other way?
6. Remember

6 steps · 0 check questions · 3 practice questions

33. Pythagoras

In a right-angled triangle, $a^2 + b^2 = c^2$, where c is the longest side, opposite the right angle.

YOU WILL LEARN TO

- Find the hypotenuse.
- Find a shorter side.
- Use Pythagoras in real problems.

BOARD STEPS

1. Squares on the sides
2. Worked example 1
3. Worked example 2
4. Worked example 3
5. Why not the other way?
6. Remember

6 steps · 0 check questions · 3 practice questions

Graphs

34. Coordinates: finding points on a grid

(x, y) means go x across, then y up. Distances between points come from Pythagoras.

YOU WILL LEARN TO

- Read and plot points (x, y).
- Find the midpoint of two points.
- Find the distance between two points.

BOARD STEPS

1. Across, then up
2. Worked example 1
3. Worked example 2
4. Worked example 3
5. Why not the other way?
6. Remember

6 steps · 0 check questions · 3 practice questions

35. Reading bar charts and tables

Before answering, read the title, the labels and the units. Then find, compare and calculate.

YOU WILL LEARN TO

- Read values from a bar chart.
- Read and total a table.
- Compare values with differences and percentages.

BOARD STEPS

1. Favourite fruit of 40 students
2. Worked example 1
3. Worked example 2
4. Worked example 3
5. Why not the other way?
6. Remember

6 steps · 0 check questions · 3 practice questions

Statistics & probability

36. The mean (average)

Mean = total ÷ how many. Total = mean × how many, which solves most average questions.

YOU WILL LEARN TO

- Find the mean.
- Find a missing value from the mean.
- Find a combined average.

BOARD STEPS

1. Levelling out
2. Worked example 1
3. Worked example 2
4. Worked example 3
5. Why not the other way?
6. Remember

6 steps · 0 check questions · 3 practice questions

37. Median and mode

Median: sort, then take the middle. Mode: the value that appears most often.

YOU WILL LEARN TO

- Find the median of odd and even lists.
- Find the mode.
- Choose mean, median or mode.

BOARD STEPS

1. Sort first, then the middle
2. Worked example 1
3. Worked example 2
4. Worked example 3
5. Why not the other way?
6. Remember

6 steps · 0 check questions · 3 practice questions

38. Sets and Venn diagrams

$n(A \cup B) = n(A) + n(B) - n(A \cap B)$. The overlap is counted twice, so subtract it once.

YOU WILL LEARN TO

- Read set notation: $\cup$, $\cap$, complement.
- Fill a Venn diagram.
- Find $n(A \cap B)$ from totals.

BOARD STEPS

1. Tea and coffee
2. Worked example 1
3. Worked example 2
4. Worked example 3
5. Why not the other way?
6. Remember

6 steps · 0 check questions · 3 practice questions

39. Population, sample and fair sampling

The population is everyone you want to know about. A sample is the part you study. A random sample gives everyone an equal chance.

YOU WILL LEARN TO

- Tell the population from the sample.
- Describe simple random sampling.
- Spot a biased sample.

BOARD STEPS

1. A sample from a population
2. Worked example 1
3. Worked example 2
4. Worked example 3
5. Why not the other way?
6. Remember

6 steps · 0 check questions · 3 practice questions

40. Probability: how likely?

Probability = ways you want ÷ all equally likely ways. It is between 0 (impossible) and 1 (certain).

YOU WILL LEARN TO

- Place events on a 0 to 1 scale.
- Find simple probabilities.
- Use P(not A) = 1 – P(A).

BOARD STEPS

1. A bag of balls
2. Worked example 1
3. Worked example 2
4. Worked example 3
5. Why not the other way?
6. Remember

6 steps · 0 check questions · 3 practice questions

Advanced: speed, work & rates

41. Speed, distance and time

Distance = speed × time. Cover the one you want in the triangle. Keep the units matching.

YOU WILL LEARN TO

- Use D = S × T in all three ways.
- Change minutes to hours.
- Change km/h to m/s.

BOARD STEPS

1. The speed triangle
2. Worked example 1
3. Worked example 2
4. Worked example 3
5. Why not the other way?
6. Remember

6 steps · 0 check questions · 3 practice questions

42. Average speed and relative speed

Average speed = total distance ÷ total time. Moving towards each other, add speeds; same direction, subtract.

YOU WILL LEARN TO

- Find the average speed of a whole journey.
- Use relative speed for meeting problems.
- Use relative speed for catching up.

BOARD STEPS

1. Two trains meeting
2. Worked example 1
3. Worked example 2
4. Worked example 3
5. Why not the other way?
6. Remember

6 steps · 0 check questions · 3 practice questions

43. Working together

If A finishes a job in 6 days, A does 1/6 of it each day. Add the daily amounts, not the days.

YOU WILL LEARN TO

- Turn "days to finish" into "part done per day".
- Find the time for two people together.
- Find one person's time from the total.

BOARD STEPS

1. Painting one wall
2. Worked example 1
3. Worked example 2
4. Worked example 3
5. Why not the other way?
6. Remember

6 steps · 0 check questions · 3 practice questions

44. Pipes and tanks

A pipe that fills a tank in 4 hours fills 1/4 of it each hour. Add the rates of filling pipes; take away the rates of outlets.

YOU WILL LEARN TO

- Find the time for two pipes together.
- Handle an outlet or a leak.
- Solve the "one pipe is faster" question.

BOARD STEPS

1. Two pipes and a leak
2. Worked example 1
3. Worked example 2
4. Worked example 3
5. Why not the other way?
6. Remember

6 steps · 0 check questions · 3 practice questions

Reasoning: patterns, logic & puzzles

45. Number and letter series

Find the rule: differences, then times, then two series mixed. A rule only counts if it fits every term.

YOU WILL LEARN TO

- Find rules using differences.
- Spot times patterns and mixed series.
- Solve letter series with positions.

BOARD STEPS

1. Look at the gaps
2. Worked example 1
3. Worked example 2
4. Worked example 3
5. Why not the other way?
6. Remember

6 steps · 0 check questions · 3 practice questions

46. Coding and decoding

Compare the word and its code letter by letter, find the rule, then use the same rule.

YOU WILL LEARN TO

- Use letter positions and EJOTY.
- Find shift and reverse codes.
- Use opposite letters.

BOARD STEPS

1. CAT becomes DBU
2. Worked example 1
3. Worked example 2
4. Worked example 3
5. Why not the other way?
6. Remember

6 steps · 0 check questions · 3 practice questions

47. All, some and none: syllogisms

Draw circles for the statements. A conclusion is "definitely true" only if it holds in every possible drawing.

YOU WILL LEARN TO

- Draw "all", "some" and "no" as circles.
- Judge conclusions from the statements only.
- Spot "cannot be determined".

BOARD STEPS

1. Three pictures
2. Worked example 1
3. Worked example 2
4. Worked example 3
5. Why not the other way?
6. Remember

6 steps · 0 check questions · 3 practice questions

48. If-then, necessary and sufficient

"If P then Q": P gives Q, and no Q means no P. But Q does not prove P.

YOU WILL LEARN TO

- Find the valid conclusions from if-then.
- Avoid reversing an if-then statement.
- Tell necessary from sufficient.

BOARD STEPS

1. If it rains, the ground is wet
2. Worked example 1
3. Worked example 2
4. Worked example 3
5. Why not the other way?
6. Remember

6 steps · 0 check questions · 3 practice questions

49. Arrangement puzzles

Draw the places, place fixed facts first, then blocks, then test each case. "Must" means true in every valid order.

YOU WILL LEARN TO

- Turn rules into short notes.
- Place fixed facts and blocks first.
- Answer must-be-true and could-be-true questions.

BOARD STEPS

1. The piano recital (7 students)
2. Worked example 1
3. Worked example 2
4. Worked example 3
5. Why not the other way?
6. Remember

6 steps · 0 check questions · 3 practice questions

50. Directions, clocks and gears

Draw directions with north at the top. Clock angle = |30H − 5.5M|. Meshing gears turn opposite ways.

YOU WILL LEARN TO

- Solve direction and distance questions.
- Find the angle between clock hands.
- Reason about gears and turns.

BOARD STEPS

1. Directions and clocks
2. Worked example 1
3. Worked example 2
4. Worked example 3
5. Why not the other way?
6. Remember

6 steps · 0 check questions · 3 practice questions